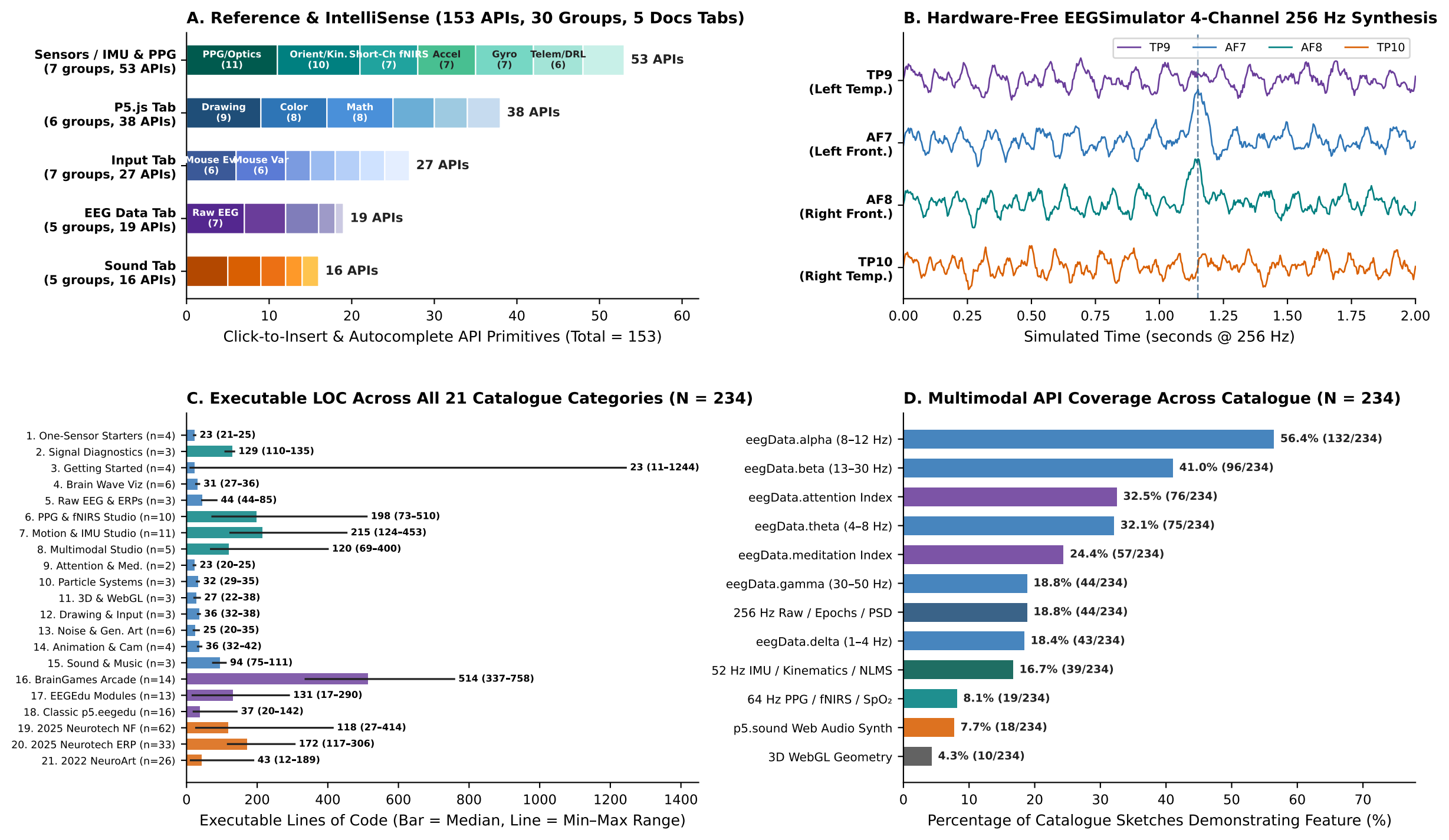

A. Reference & IntelliSense (153 APIs, 30 Groups, 5 Docs Tabs)
Sensors / IMU & PPG
(7 groups, 53 APIs)
PPG/Optics (11)
Orient/Kin. (10)
Short-Ch fNIRS (7)
Accel (7)
Gyro (7)
Telem/DRL (6)
53 APIs
P5.js Tab
(6 groups, 38 APIs)
Drawing (9)
Color (8)
Math (8)
38 APIs
Input Tab
(7 groups, 27 APIs)
Mouse Ev (6)
Mouse Var (6)
27 APIs
EEG Data Tab
(5 groups, 19 APIs)
Raw EEG (7)
19 APIs
Sound Tab
(5 groups, 16 APIs)
16 APIs
0
10
20
30
40
50
60
Click-to-Insert & Autocomplete API Primitives (Total = 153)
B. Hardware-Free EEGSimulator 4-Channel 256 Hz Synthesis
TP9
AF7
AF8
TP10
TP9
(Left Temp.)
AF7
(Left Front.)
AF8
(Right Front.)
TP10
(Right Temp.)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Simulated Time (seconds @ 256 Hz)
C. Executable LOC Across All 21 Catalogue Categories (N = 234)
1. One-Sensor Starters (n=4)
23 (21–25)
2. Signal Diagnostics (n=3)
129 (110–135)
3. Getting Started (n=4)
23 (11–1244)
4. Brain Wave Viz (n=6)
31 (27–36)
5. Raw EEG & ERPs (n=3)
44 (44–85)
6. PPG & fNIRS Studio (n=10)
198 (73–510)
7. Motion & IMU Studio (n=11)
215 (124–453)
8. Multimodal Studio (n=5)
120 (69–400)
9. Attention & Med. (n=2)
23 (20–25)
10. Particle Systems (n=3)
32 (29–35)
11. 3D & WebGL (n=3)
27 (22–38)
12. Drawing & Input (n=3)
36 (32–38)
13. Noise & Gen. Art (n=6)
25 (20–35)
14. Animation & Cam (n=4)
36 (32–42)
15. Sound & Music (n=3)
94 (75–111)
16. BrainGames Arcade (n=14)
514 (337–758)
17. EEGEdu Modules (n=13)
131 (17–290)
18. Classic p5.eegedu (n=16)
37 (20–142)
19. 2025 Neurotech NF (n=62)
118 (27–414)
20. 2025 Neurotech ERP (n=33)
172 (117–306)
21. 2022 NeuroArt (n=26)
43 (12–189)
0
200
400
600
800
1000
1200
1400
Executable Lines of Code (Bar = Median, Line = Min–Max Range)
D. Multimodal API Coverage Across Catalogue (N = 234)
eegData.alpha (8–12 Hz)
56.4% (132/234)
eegData.beta (13–30 Hz)
41.0% (96/234)
eegData.attention Index
32.5% (76/234)
eegData.theta (4–8 Hz)
32.1% (75/234)
eegData.meditation Index
24.4% (57/234)
eegData.gamma (30–50 Hz)
18.8% (44/234)
256 Hz Raw / Epochs / PSD
18.8% (44/234)
eegData.delta (1–4 Hz)
18.4% (43/234)
52 Hz IMU / Kinematics / NLMS
16.7% (39/234)
64 Hz PPG / fNIRS / SpO₂
8.1% (19/234)
p5.sound Web Audio Synth
7.7% (18/234)
3D WebGL Geometry
4.3% (10/234)
0
10
20
30
40
50
60
70
Percentage of Catalogue Sketches Demonstrating Feature (%)